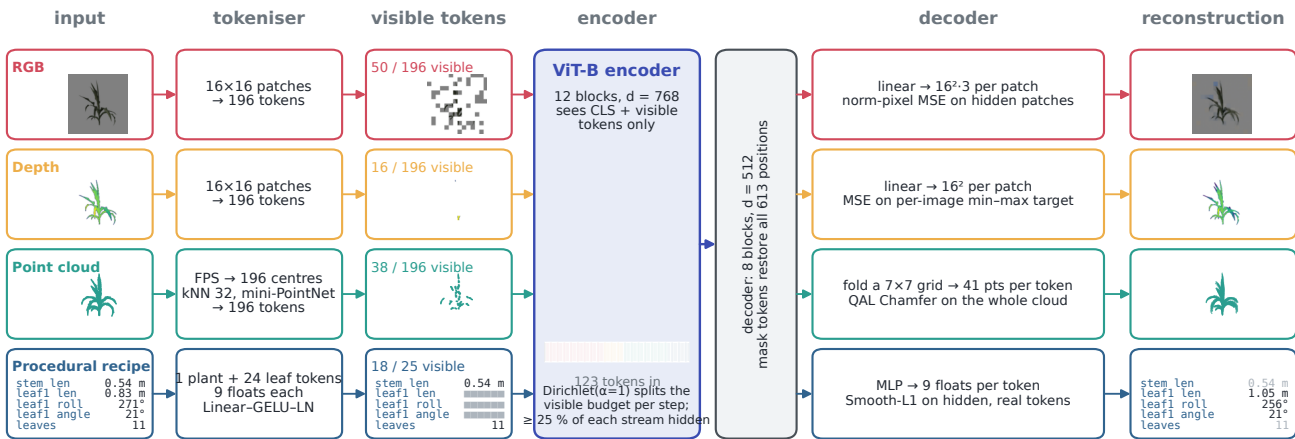
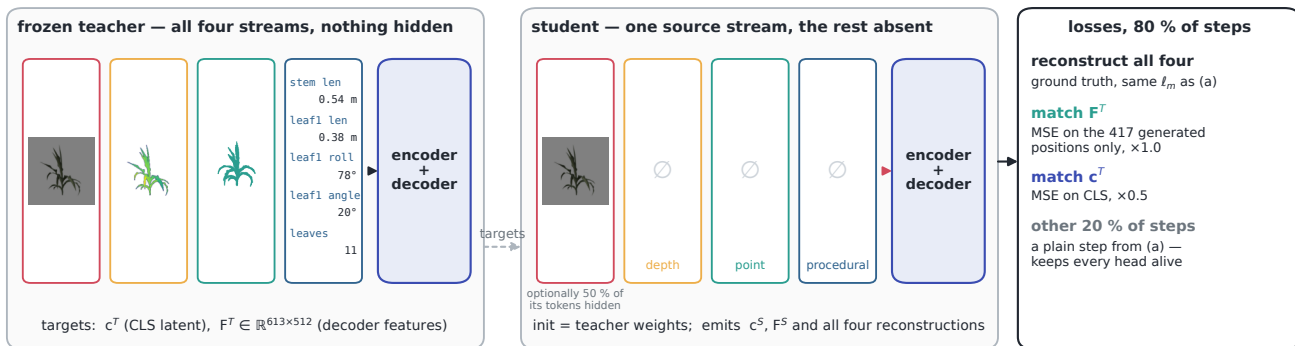


(a) Four-stream masked pretraining



The decoder layout never depends on the mask, so two forward passes with different visible sets give position-aligned features — the hook for (b).

(b) A vision-complete teacher and single-sensor students



all panels: held-out plant Sorghum_10_04, real inputs, masks and model outputs; grey recipe values in (a) are visible tokens passed through, not predictions